

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 302 — Past Questions & Materials
by Renance

Course: PEE 302
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — PEE302

Level 300 • Second Semester • PastQuestion

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QUESTION 3 (16 marks)

3(a) A modular refinery needs N50,000,000 for a turnaround maintenance project in 6 years.
 (i) If it sets up an account to save up the capital, at an interest rate of 9% per annum, for 6 years, calculate the required annual deposit.
 (ii) If the capital is borrowed from a bank at 5% interest rate/month to be repaid over the same 6 years, calculate the monthly repayment.

3(c) State five factors which determine the price of crude oil and briefly explain each of the following factors.

QUESTION 4 (16 marks)

4(a) Determine the mean, variance and standard deviation of the proposed exploratory well shown below.

$\sum P_i [X^2 + (n-1)X]$
 $P_1 = 0.30$

0 (bbl) 200 (bbl) 250 (bbl) 400 (bbl)

4(b) List and explain four categories of risk that affect economic viability of a project.
 4(c) What is the balance in an account at the end of 7 years, if \$18,200 is deposited today and the account earns 6% interest, compounded annually and quarterly

QUESTION 5 (16 marks)

5(a) Explain the following terms (i) Royalty (ii) Signature bonus (iii) Fiscal term (iv) Tax (v) Profit sharing.
 5(bi) Briefly discuss the Nigerian oil Concession & list five features of the early Concessions
 (bii) State four (4) Significance of Concessions
 5(c) Determine the effective interest rate, given that the Nominal interest rate is 18% (Tabulate your readings)

m	I(%)
1	18.00
2	9.00
4	4.50
12	1.5

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF PETROLEUM ENGINEERING

SECOND SEMESTER EXAMINATIONS 2025/2026 ACADEMIC SESSION

FUPRE PEE 302: BASIC PETROLEUM ECONOMICS

Answer Question 1 and any other three (3)

Time: 3 Hours

Instruction: Don't write on your question paper.

QUESTION 1 (22 Marks)

1(a) An investment produced a series of cash inflows below with an initial investment of \$60,000 at 10%, 15%, 20%, 25%, 30%, interest rates calculate.

- (i) The NPV at various interest rates..
- (ii) Graphically determine the internal rate of return (IRR).
- (ii) The payback period at 12 % (PB)
- (iii) The profitability index at 12% interest rate

Cash Flows (\$)	37,100	16,800	12,200	8,640	5,440	2500
Time (years)	1	2	3	4	5	6

1(b) Mention three tools use for evaluating investment and state two importance of each investment tools.

QUESTION 2 (16 marks)

2(a) Define the following terms:

- (i) Capital Expenditure (ii) Unit Technical Cost, (iii) Abandonment Cost (iv) Operating Cost

2(b) The Present value to buy a single pipe to run a maintenance operation along a flowline is \$9,600. What is the Future Value to buy the same pipe after 10 years at a 9% interest rate per year?

2(c) A Company is planning to drill 10 exploratory wells with an estimated chance of 20% success. Using Binomial distribution to determine the probability that:

- (i) The drilling operation will produce exactly five wells?
- (ii) The drilling operation will produce at least three successful wells?
- (iii) The drilling operation will produce at most four successful wells?